

Asad Hayat

✉ Email address: asadhayat@stu.tyust.edu.cn

📍 Home: (China)

EDUCATION AND TRAINING

Doctor of Philosophy in Control Science

Taiyuan University of Science and Technology [10/2023 – Current]

City: Taiyuan | Country: China

Master of Science in Computer Science

Capital University of Science and Technology Islamabad [09/2018 – 04/2021]

City: Islamabad | Country: Pakistan

CGPA: 3.24/4.00

Thesis: A Load-Balanced Task Scheduler for Heterogeneous Systems Based on Machine Learning [link](#).

Short Description: This master thesis focus on designing a task scheduler that distributes programs among the processing devices through machine learning. The model is implemented on OpenCL programs from two benchmark suite, Polybench and AMD in heterogeneous environment. At first the execution time of program is predicted on both CPU and GPU device through machine learning and then the program is assigned to a processing device through the designed scheduler. The scheduler reduces the overall execution time of programs and utilizes the processing devices as compare to base line scheduling schemes.

Bachelor of Science in Information Technology

IBMS, The University of Agriculture Peshawar [01/2014 – 02/2018]

City: Peshawar | Country: Pakistan

CGPA: 3.47/4.00

Final Year Project: MS Management System

Short Description: This desktop application was designed to keep track of the record of all the Master students of IBMS, The University of Agriculture Peshawar. A separate profile was created for each Master Student and all the documents that he/she submit to university is stored in soft form. The application was designed in Java programming language in connection with SQL server.

RESEARCH INTERESTS

Computational Intelligence, Evolutionary Algorithms, Machine Learning, Deep Learning, Data Mining, Data Science

WORK EXPERIENCE

Computer science lecturer

Lahore Leads University [11/2021 – 09/2023]

City: Lahore | Country: Pakistan

- Teaching Subjects like Introduction to Computer and Introduction to Data Science.
- Organizing labs for Introduction to Computer Subject.

RESEARCH EXPERIENCE

[10/2023 – Current]

Research Associate

Working on Optimization at Data Science and Computational Intelligence Lab, Taiyuan University of Science and Technology, Taiyuan.

- Studying and researching about Surrogate Assisted Evolutionary Algorithms.
- Focus on Designing Single and Multi-Objective Optimization Algorithms.

[07/2020 – 02/2021]

Research Assistant

Working on Optimization and Utilization of CPU and GPU at CORDE Lab, Capital University of Science and Technology, Islamabad.

- Studying and Analyzing CPU's and GPU's along with OpenCL Applications.
- Designed A scheduler through machine learning for mapping applications on CPU and GPU in Optimized way.

PUBLICATIONS

Journal Publications

1. Hasnain, M., Ghani, I., Smith, D., Asaf, M., Saleem, S., **Hayat, A.**, & Jeong, S. R. (2024). Recurrent Neural Network (RNN)-Based Approach to Uncover the Relationship Between Block Size and Blockchain Performance.
2. Nadir, M. N., Rathore, M. S., **Hayat, A.**, & Mansoor, J. A. (2024). CPU vs. GPU: Performance comparison of OpenCL Applications on a Heterogeneous Architecture. *Journal of Computing & Biomedical Informatics*, 7(02).
3. **Hayat, A.**, Khalid, Y. N., Rathore, M. S., & Nadir, M. N. (2023). A machine learning-based resource-efficient task scheduler for heterogeneous computer systems. *The Journal of Supercomputing*, 1-29.
4. Hasnain, M., **Hayat, A.**, & Hussain, A., Revolutionizing Chronic Obstructive Pulmonary Disease Care with the Open AI Application: ChatGPT. *Ann Biomed Eng* (2023).
5. Sajawal, M., Sardar, U., Hamed, S. A., **Hayat, A.**, and Ashraf, M. U. (2022). A Predictive Analysis of Retail Sales Forecasting Using Machine Learning Techniques *Lahore Garrison University Research Journal of Computer Science and Information Technology*, 6(04), 23–33.

Conference

1. **A Hayat**, MN Nadir, MS Rathore, (2022) A Load-Balanced Task Scheduler for Heterogeneous Systems based on Machine Learning *1st International Conference on Computing Research (ICCoR 22)*, 13-14
2. MN Nadir, **A Hayat**, MS Rathore, (2022) Analysis of OpenCL Applications on CPU and GPU *1st International Conference on Computing Research (ICCoR 22)*, 56-57

ACADEMIC RESEARCH AND DEVELOPMENT PROJECTS

Visualizing Public and Private Universities of Pakistan

Fall 2019

Research Project in which All the public and private universities of Pakistan are visualized through different aspects like all universities of Pakistan, universities of each province, departments of a university etc. [Project Link](#)

Sentiment analysis of Conspiracy Theories about Covid-19 through Machine Learning.

Sentiment analysis using English-language tweets obtained online. The tweets were retrieved, preprocessed, and analyzed using text mining techniques, followed by sentiment analysis using machine learning algorithms. [Link](#)

DATA SCIENCE SKILLS

Programming Languages

MATLAB, Python, Java, C/C++, Latex.

Data Science

Machine learning, Artificial Intelligence, Deep learning, NLP.